

Generating Intermediate Representation (IR) .txt format from DFG IR

We take the example of **8Point DCT** to explain the IR file creation process from its DFG IR. The generic function (high-level code) of 8-Point DCT to compute a sample o/p is [1][2][3]:

$$X[0] = k1 * x[0] + k2 * x[1] + k3 * x[2] + k4 * x[3] + k5 * x[4] + k6 * x[5] + k7 * x[6] + k8 * x[7]$$

The respective DFG is shown below [1][2][3]:

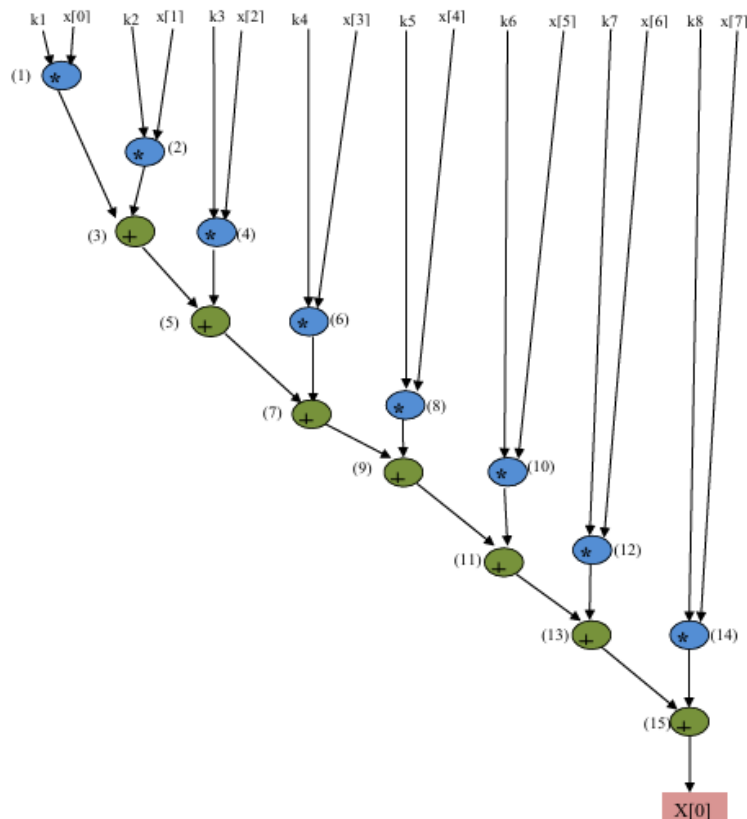
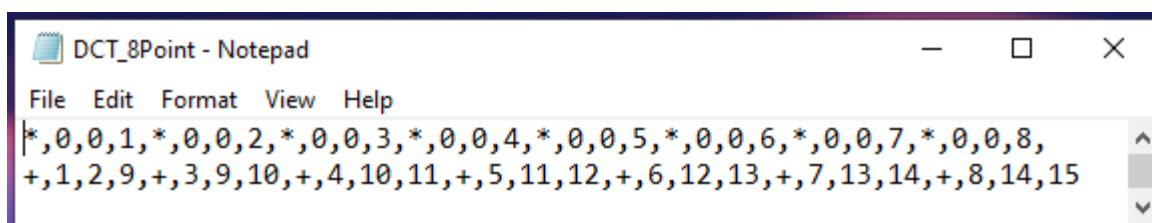


Fig. Data flow graph of an 8-point DCT

The respective .txt file format for the above DFG IR is shown below:



Similarly, other DFG IRs have been converted into its respective .txt file format (IR).

The HLS Benchmarks in the form of high-level description (transfer function) and its associated intermediate representation (IR) as CDFG/DFG is provided in the "HLS Benchmarks.pdf", also publicly available in:

<https://www.anirban-sengupta.com/HLS%20Benchmarks%20Details.pdf>

The IR in the form of text file format (such as DCT_8Point.txt, FIR.txt etc) is generated from the DFG above.

Note: The relevant research paper citations, where the high-level description (transfer function) and its associated intermediate representation (IR) as CDFG/DFG is described, is also present in the PDF file.

References:

- [1] A. Sengupta, R. Chaurasia and A. Anshul, "Robust Security of Hardware Accelerators Using Protein Molecular Biometric Signature and Facial Biometric Encryption Key," **IEEE Transactions on Very Large Scale Integration (VLSI) Systems**, vol. 31, no. 6, pp. 826-839, June 2023, doi: 10.1109/TVLSI.2023.3265559.
- [2] A. Sengupta "High-Level Synthesis based Methodologies for Hardware Security, Trust and IP Protection, " **The Institute of Engineering and Technology (IET)**, 2024, ISBN-13: 978-1-83724 117-0.
- [3] A. Sengupta, S. P. Mohanty "IP Core Protection and Hardware-Assisted Security for Consumer Electronics", **The Institute of Engineering and Technology (IET)**, 2019, Book ISBN: 978-1-78561-799-7, e-ISBN: 978-1-78561-800-0.